

Predicting Patent Examination Pendency with Predictive Process Monitoring

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Abstract. Patent pendency affects applicant costs, examiner resource allocation, and the timing of innovation commercialization. This study models USPTO patent examination as an event log and predicts remaining time to grant from partial prosecution histories. We compare three feature-richness levels: process-only features, intra-case patent attributes, and inter-case resource-related features. Using filing-to-grant traces from the USPTO Patent Examination Research Dataset, we find that richer contextual information improves prediction accuracy. For XG-Boost, mean absolute error decreases from 214.8 days with process-only features to 211.4 days with intra-case features, and further to 204.3 days with inter-case features. These results suggest that patent pendency is not only a function of an application’s own prosecution history, but also of the broader examination environment, including examiner workload and art-unit backlog. The study demonstrates the value of predictive process monitoring for patent examination and highlights the predictive role of contextual information.

Keywords: Remaining time prediction · Patent examination · USPTO.

1 Introduction

Patent pendency affects applicant cost, commercialization timing, and examiner resource allocation. Delays in patent issuance can reduce the window available to commercialize a technology, affect licensing and investment decisions, and place applicants in less favorable bargaining positions during technology transfer [2]. Despite the practical importance of grant timing, prior work has largely treated patent examination as a static prediction problem. In this study, we apply predictive process monitoring to USPTO patent examination records, treating each application as a case, each transaction as an event, and the task is to predict the remaining time to grant from an incomplete prosecution history. Our goal is not only to predict pendency, but also to examine whether richer contextual information improves prediction accuracy. This study makes three contributions. First, we construct filing-to-grant event traces from raw USPTO administrative records. Second, we formulate patent pendency prediction as a prefix-based remaining-time regression task. Third, we compare three feature-richness levels: process-only features, intra-case patent attributes, and inter-case resource-aware features.

2 Related Work

Prior studies on patent pendency use application-level information such as technology field, applicant characteristics, and claim-related variables to predict examination duration or grant status [3]. While useful for identifying factors associated with pendency, these approaches do not capture how the expected remaining duration changes as an application progresses through office actions and applicant responses.

A related stream models patent examination as an event sequence. For example, Wu et al. [1] use USPTO examination logs to learn process-aware representations for multiple patent prediction tasks. This line of work shows that the prosecution history itself contains useful information beyond static application characteristics. Our study shares this process view, but focuses specifically on remaining-time prediction from partial prosecution histories. Rather than learning a general-purpose representation, we examine how different types of features contribute to pendency prediction.

Remaining-time prediction is a central task in predictive process monitoring (PPM). Prior PPM studies show that event histories, trace encodings, and inter-case information such as workload or congestion can improve prediction accuracy [4, 5]. These insights are especially relevant for patent examination because applications are processed within a shared examination system, where examiner workload and art-unit congestion may affect the pace of prosecution. Building on this literature, we compare process-only, intra-case, and inter-case feature sets under the same prefix-based prediction framework.

3 Patent Examination as an Event Log

Data. We use the USPTO Patent Examination Research Dataset (PatEx), which contains administrative transaction records for patent applications. Each record includes an application number, a transaction date, and a text description of the event. In our setting, each patent application is treated as one case, the transaction date is treated as the timestamp, and the event description is treated as the activity. We also use application- and patent-level information to construct additional features. These include examiner art unit, patent type, number of claims, and inventor count. The application-level information is merged with the event log by application number, while patent-level features are merged using the patent number.

From raw USPTO records to event logs. Table 1 summarizes how we convert the raw USPTO administrative records into an event log. The case identifier is the application number, so each patent application corresponds to one process instance. The activity label is the text description of each transaction event, and the timestamp is the recorded date of the transaction.

We sort events within each application by the recorded date and then define a filing-to-grant trace by keeping events from the first filing receipt to the first

Table 1. Mapping raw USPTO records to event-log elements.

Element	Raw field	Definition
Case ID	<code>application_number</code>	Patent application
Activity	<code>event_desc</code>	Text description of a transaction event
Timestamp	<code>recorded_date</code>	Event date
Start	<code>event_desc</code>	First Filing Receipt
End	<code>event_desc</code>	First grant-mail record
Attributes	application/patent tables	Art unit, claims, inventors

Table 2. Final event-log sample and prefix-level temporal split.

Sample	Cases	Rows
Full analysis sample	1,452,372	102,304,746
Training set	982,721	66,724,046
Test set	469,651	33,087,963

grant record. This step converts the raw administrative transaction records into process traces that can be used for prefix-based remaining-time prediction.

Preprocessing the administrative log. The raw USPTO log contains both substantive prosecution events and administrative records. Because these different types of records are mixed together, the raw log does not directly define the filing-to-grant process that we want to analyze.

We restrict the analysis sample to granted utility patent applications filed between 2008 and 2015. The lower bound reflects the introduction of electronic filing at the USPTO around 2008, after which **Filing Receipt** appears consistently across applications and the event log is more uniformly structured. The upper bound addresses a potential selection bias: including more recent applications would exclude cases still under examination, leaving only applications that were granted quickly and thereby distorting the pendency distribution.

We further restrict the sample to applications that contain a **Recordation of Patent Grant Mailed** event, because the raw event log includes post-grant records such as assignment transactions, maintenance fee payments, and patent expiration. By anchoring the trace end at its first occurrence, all traces share a common endpoint.

As a result, all traces in the final sample share a common structure: they begin with **Filing Receipt** and end with **Recordation of Patent Grant Mailed**. After applying these filtering and trace construction rules, the final sample contains 1,452,372 granted applications and 102,304,746 event rows. We apply a case-level temporal split based on the trace start date: applications beginning before 2014 form the training set, and those beginning in 2014 or later form the test set (Table 2).

Table 3. The three feature configurations. Each level extends the previous one.

Approach	Adds	Features	#
M1 Process-only	Event history	current activity, event order, elapsed days	3
M2 + Intra-case	Patent attributes	examiner art unit, no. claims, no. inventors	+3
M3 + Inter-case	Resource context	examiner workload, art-unit backlog	+2

4 Method

Prefix extraction and target definition. From each application trace $\sigma = \langle e_1, \dots, e_n \rangle$, we construct prefix-level prediction samples. Each non-terminal event becomes one prediction point. At prefix k , the model observes information available up to event e_k and predicts the remaining number of days until grant:

$$r_k = t_n - t_k, \quad (1)$$

where t_k is the date of the current event and t_n is the date of the grant endpoint. The terminal grant event is excluded because its remaining time is zero. For each prefix, we also compute the elapsed time since the beginning of the trace:

$$elapsed_k = t_k - t_1, \quad (2)$$

where t_1 is the date of the filing receipt.

Feature-richness levels. We compare three feature configurations. Each level extends the previous one, while the train-test split and model settings are held fixed. The process-only model uses a last-state encoding of the current event. It includes the current event description, the event order within the trace, and the number of elapsed days since filing receipt. The intra-case model adds static patent-level attributes: examiner art unit, number of claims, and inventor count. The inter-case model further adds resource-aware features: examiner workload and art-unit backlog.

Computing inter-case features. Examiner workload and art-unit backlog are computed from the full PatEx event log. For each examiner (art unit) and calendar month, we count the number of applications with an active span covering that month. Each prefix is matched to the counts from its event month.

Train-test split and leakage control. As described above, we use a case-level temporal split based on the trace start date. All prefixes from the same application remain on the same side of the split, so the same case does not appear in both training and test data.

We also avoid using information that would not be available at the time of prediction. Categorical encoders are fit using only the training data, and categories that appear only in the test set are assigned to a default class. Inter-case features are measured at the month of the current event rather than using later information.

Table 4. Pendency prediction error by model and feature-richness level.

Model	Process-only		Intra-case		Inter-case	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
Baseline	305.7	384.1	–	–	–	–
Linear Reg.	291.2	363.9	289.6	362.4	277.7	351.6
XGBoost	214.8	310.0	211.4	302.2	204.3	294.2

5 Experiments and Results

Experimental setup. We compare three models: a mean-prediction baseline, linear regression, and XGBoost. The mean baseline predicts the average remaining time in the training data. Linear regression provides a simple parametric benchmark. XGBoost is used as a non-linear tree-based model that can capture interactions among event, case, and resource features. For XGBoost, we use 100 trees, a maximum depth of 6, and a learning rate of 0.1. The same train-test split and model settings are used across the three feature configurations. Prediction accuracy is evaluated using mean absolute error (MAE) and root mean squared error (RMSE), both measured in days.

Results. Table 4 shows the test errors for each model and feature configuration. Overall, the results show two main patterns. First, model choice matters. XGBoost performs better than linear regression and the mean baseline across all feature levels. With process-only features, XGBoost reduces MAE from 305.7 days in the mean baseline to 214.8 days. Second, adding richer contextual information improves prediction accuracy. For XGBoost, MAE decreases from 214.8 to 211.4 and further to 204.3 days across the three feature levels, with RMSE following the same pattern. These results suggest that patent pendency is related not only to the observed history of an individual application, but also to application-level attributes and the broader examination environment. In particular, the improvement from inter-case features indicates that examiner workload and art-unit backlog provide additional predictive value beyond the prosecution history itself.

6 Discussion and Limitations

The results suggest that patent pendency prediction improves when the model uses contextual information beyond the observed event sequence. Process-only features capture where an application is located in its own prosecution history. However, adding intra-case features allows the model to account for differences across applications, and adding inter-case features further captures the examination environment, such as examiner workload and art-unit backlog.

This is important because patent pendency is not only a within-case process. Applications with similar prosecution histories may still have different remaining

times depending on the technology area, examiner, or congestion in the relevant art unit. The improvement from the inter-case model supports this view, indicating that resource-related context provides additional predictive value beyond the case history itself.

This study has several limitations. First, we restrict the sample to granted applications because remaining time to grant is observed only for these cases. As a result, the findings should be interpreted as predictions for applications that eventually receive a grant, rather than for all filed applications. Second, our inter-case features are simple proxies for examination congestion. Examiner workload and art-unit backlog capture part of the resource environment, but they do not fully measure examiner expertise, applicant behavior, claim complexity, or policy changes at the USPTO.

Finally, this study focuses on prediction rather than causal interpretation. The results show that contextual features are useful for improving prediction accuracy, but they do not imply that workload or backlog causally determines pendency. Future research could develop richer time-varying features and combine predictive process monitoring with causal designs to study how changes in examination resources affect patent pendency.

7 Conclusion

This study modeled USPTO patent examination as an event log and compared process-only, intra-case, and inter-case feature sets for remaining-time prediction. The results show that richer contextual information improves prediction accuracy, and that examiner workload and art-unit backlog provide predictive value beyond the prosecution history itself. Future work can extend the analysis to non-granted applications and examine the causal role of examination resources in patent pendency.

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